

Criterio de Dini

Proposición 1 (Criterio de Dini). Sea $f \in \mathcal{L}^1([0, 1))$ 1-periódica y $x_0 \in (-1/2, 1/2)$ tal que $f(x_0) < \infty$. Entonces,

$$\exists \delta > 0 : \int_{|t| < \delta} \left| \frac{f(x_0 + t) - f(x_0)}{t} \right| dt < \infty \implies S_N(f)(x_0) \xrightarrow{N \rightarrow \infty} f(x_0).$$

Demostración: Puesto que $\forall N \in \mathbb{N} : \int_{-1/2}^{1/2} D_N(t) dt = 1$ (obs-integral-nucleo-dirichlet/Observac

$$\begin{aligned} S_N f(x_0) - f(x_0) &= \int_{-1/2}^{1/2} f(x_0 - y) D_N(y) dy - f(x_0) \int_{-1/2}^{1/2} D_N(y) dy \\ &= \int_{-1/2}^{1/2} [f(x_0 - y) - f(x_0)] D_N(y) dy \\ &\stackrel{\text{lem-formula-nucleo-dirichlet/??}}{=} \int_{-1/2}^{1/2} [f(x_0 - y) - f(x_0)] \frac{\sin((2N+1)\pi y)}{\sin(\pi y)} dy. \end{aligned}$$

Definimos $I_N f(x_0)$ como la integral anterior en el intervalo $|y| < \delta$ y $J_N f(x_0)$ como la integral en el intervalo $\delta \leq |y| \leq 1/2$. Entonces $S_N f(x_0) - f(x_0) = I_N f(x_0) + J_N f(x_0)$.

$$\begin{aligned} J_N f(x_0) &= \int_{\delta \leq |y| \leq 1/2} [f(x_0 - y) - f(x_0)] \frac{e^{\pi i(2N+1)y} - e^{-\pi i(2N+1)y}}{2i \sin(\pi y)} dy \\ &= \int_{-1/2}^{1/2} h_{+,x_0}(y) e^{2\pi i N y} dy - \int_{-1/2}^{1/2} h_{-,x_0}(y) e^{-2\pi i N y} dy, \end{aligned}$$

donde $h_{\pm, x_0}(y) = [f(x_0 - y) - f(x_0)] \frac{e^{\pm \pi i y}}{2i \sin(\pi y)} \mathbb{1}_{\{\delta \leq |y| \leq 1/2\}}(y)$. Entonces, tenemos que

$$J_N f(x_0) = \widehat{h_{+,x_0}}(-N) - \widehat{h_{-,x_0}}(N) \xrightarrow{N \rightarrow \infty} 0$$

por el Lema de Riemann-Lebesgue. Solo falta ver que $h_{\pm, x_0} \in \mathcal{L}^1([-1/2, 1/2])$:

$$\int_{-1/2}^{1/2} |h_{\pm, x_0}(y)| dy = \int_{\delta \leq |y| \leq 1/2} |f(x_0 - y) - f(x_0)| \frac{1}{2 |\sin(\pi y)|} dy.$$

Como $\forall y \in [-1/2, 1/2] : \delta \leq |y| \leq 1/2 \implies |\sin(\pi y)| \geq |\sin(\pi \delta)| > 0$, tenemos que

$$\|h_{\pm, x_0}\|_1 \leq \frac{1}{2 |\sin(\pi \delta)|} \int_{-1/2}^{1/2} |f(x_0 - y) - f(x_0)| dy \leq \frac{1}{2 |\sin(\pi \delta)|} 2 \|f\|_1 < \infty.$$

Por otro lado, para $I_N f(x_0)$, tenemos que

$$\begin{aligned} I_N f(x_0) &= \int_{|y|<\delta} [f(x_0 - y) - f(x_0)] \frac{e^{\pi i(2N+1)y} - e^{-\pi i(2N+1)y}}{2i \sin(\pi y)} dy \\ &= \int_{-1/2}^{1/2} g_{+,x_0}(y) e^{2\pi i N y} dy - \int_{-1/2}^{1/2} g_{-,x_0}(y) e^{-2\pi i N y} dy, \end{aligned}$$

donde $g_{\pm,x_0}(y) = [f(x_0 - y) - f(x_0)] \frac{e^{\pm \pi i y}}{2i \sin(\pi y)} \mathbb{1}_{\{|y|<\delta\}}(y)$. Nuevamente, por el Lema de Riemann-Lebesgue, tenemos que $I_N f(x_0) \xrightarrow{N \rightarrow \infty} 0$ siempre que $g_{\pm,x_0} \in \mathcal{L}^1$. Pero esto es cierto ya que

$$\|g_{\pm,x_0}\|_1 = \int_{|y|<\delta} |f(x_0 - y) - f(x_0)| \frac{1}{2 |\sin(\pi y)|} dy.$$

Como $\forall y \in (-\delta, \delta) : |\sin(\pi y)| \geq |y|/2$, tenemos que

$$\|g_{\pm,x_0}\|_1 \leq \int_{|y|<\delta} \frac{|f(x_0 - y) - f(x_0)|}{|y|} dy.$$

Esta integral es finita ya que f satisface el criterio de Dini en x_0 . ■

Referenciado en

- `prop-clase-ck-velocidad-convergencia-uniforme-fourier`
- `ejem-continuidad-derivadas-laterales-imp-dini`
- `prop-criterio-dirichlet`

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